Crime & Immigration in the European Union

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Project Definition

- **TITLE:** CRIME & IMMIGRATION IN THE EU
- **DESCRIPTION:** In the current geopolitical climate, a question of immigration had become a stepping stone in propagation of different ideologies. People tend to believe statistics they have not seen and bias to the worldview that was put onto them. Because of that, we would like to propose a reliable visual source of information, to allow general public to use it and form their beliefs based on the truthful data.

  In this project we will propose a visual representation of yearly crime and immigration rate in EU.

Project Definition

➔ **DOMAIN:**

WHAT?

The domain of interest lies in analysis and visual exploration of crime statistics and immigration patterns within the European Union. It consists of understanding how crime rates evolve over time and how these patterns relate to the nationalities of individuals convicted. The domain is based on the data transparency and unbiased perceptions.

WHY?

- How did the crime & immigration rates in EU changed over the past years (from 2018 until 2022)?
- **Is there any correlation in between immigration and crime rates?**

Project Definition

➔ **DOMAIN:**

WHO and HOW?

The main stakeholders are:

- General public: seeking an unbiased information/statistics.
- Politicians: pushing certain narrative based on the statistical data.
- Journalists and media outlets: reporting about the crime and immigration.
- Researchers: using statistics for the research purposes in immigration and crime.

➔ **DATA SOURCES:**

The two main data sources are:

UN: *https://dataunodc.un.org/*

Eurostat: *https://ec.europa.eu/eurostat/en/*

Selection and Data Type

The dataset selected contains:

- ***1-dimensional:*** numerical values (number of crimes committed per 100,000 inhabitants, immigration rates, etc.) and categorical variables (crime type, nationality, country, etc.).

- ***2-dimensional geographic:*** EU countries identified by their geographic location and regions.

- ***temporal***: yearly records of crimes committed and immigration statistics.

Problem characterization - Why will the data be used?

Chosen data is going to be used to visually test and explore the relationships between crime, immigration, and population across Europe from 2018 to 2022 with the next goals:

- Identify trends over time
- Visually identify patterns without a biased narrative
- Compare all European countries throughout these years

Problem characterization

- What will the data be used for?

Is the crime rate on average in Europe grows or decreases?

Is there a relation between immigration and amount of crimes?

Is the amount of crimes in Europe grows or decreases?

~~What is the difference in crime rate between countries with the highest and the lowest ratio of immigrants?~~

~~What was the safest country throughout this period?~~

~~What is the most safe country lately?~~

~~Does bigger population imply higher crime rate?~~

~~Does geographical situation affect crime rate?~~

What is the least safe country in Europe lately?

~~In which year each country had the lowest crime rate?~~

Problem characterization

- Who will use the data?

The data, as mentioned previously in project definition, will be mainly used by general public, since it is interested in testing political narratives around crime and immigration, as well as viewing unbiased statistics about it. All sides of the political spectrum are included in the general public, but if specifying:

- General public: seeking an unbiased information/statistics with an interest in safety of the country they want to travel or migrate to.
- Politicians: pushing certain narrative based on the statistical data.
- Journalists and media outlets: reporting about the crime and immigration.
- Researchers: using statistics for the research purposes in immigration and crime.

- How will the data be used?

The data is going to be used for creating correlations between crime, immigration and populations, which will be used in graphs, visualizations, maps and textual descriptions for the ease of its comprehension.

Problem characterization

- When will the data be used?

The data we are working with can be used at any time and will hold the same importance. Now, however, it holds a crucial value due to the current political climate, which leads to an increase of immigration and pushes certain narrative into the mass.

Proposed data can also be used by travellers and immigrants themselves, who are interested in overall safety and crime statistics of the country they are visiting/moving to.

Problem characterization

- Where will the data be used?

The data will be used in an online, publicly accessible application designed for global use. Although the dataset focuses exclusively on European Union crime and immigration statistics, the platform does not restrict users from other locations from accessing it.

The interface and the data presented will remain the same regardless of where the user is located. This ensures consistent access to information and supports transparency across different regions.

Problem characterization

- Which ethical values will be promoted by the visual application?

One of our main goals is to show an unbiased statistics without any assumptions being made, as the topic of correlation of crime and immigration is highly divisive. Thus, we are eager to promote next ethical values:

- ***Transparency:*** as it was said before, the end goal of our application lies in showing the transparent and unbiased data, that will allow users to come to their own conclusions, rather than believing a certain political narrative.
- ***Trustability:*** all of the data used has been taken from official and reliable resources, in order to avoid any possible bias or misinformation.
- ***Neutrality:*** the application presents data without reinforcing any stereotypes or further relations that are not supported by evidence.
- ***Inclusivity and Accessibility:*** an application developed should be opened to all users worldwide, regardless of location and technical or educational background.

To conclude everything above, the end goal of our website lies in showing pure statistics that can be deduced from the open data sources, that can be further be used by the users in creation of their own opinions and world views.

Problem characterization - Assumption Reversal

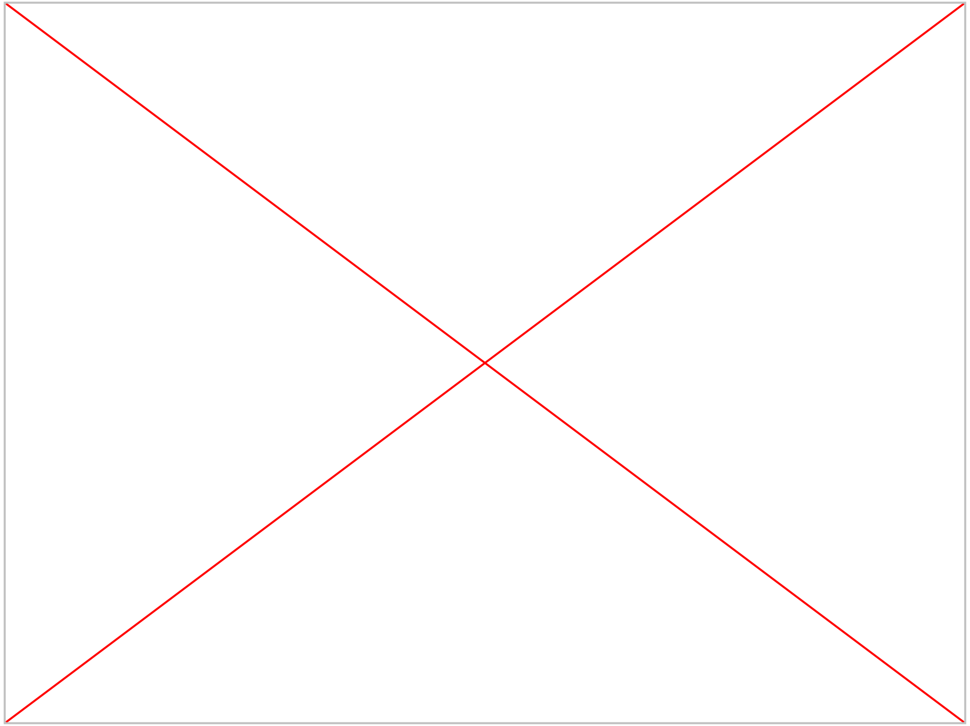
Assumption: People who cannot see the data will rely on an alternative text to voice description of the graphs.

Reversal: While text to voice descriptions can be very informative, in order to accommodate a wider population (inclusivity), we have decided to implement a real life prototype that will illustrate data both physically and textually.

So, the prototype will consist of 3D model of European Union map, with each country being separated from the other by border and type of terrain. We imply that the higher immigration rate is in a country, the higher terrain it has (also can be thought of as a mountain that starts at the borders of a country). On the other hand, the higher crime rate is in a country, the more vibration surface of this country will have.

This covers both inclusivity and universal design of a prototype, allowing all of the users to understand and get knowledge from our system.

Problem characterization - Assumption Reversal



Definition of operations / tasks

We have defined those tasks to answer the pointed out questions considering the following:

- ➔ ***overview***: there is no overview of the dataset
- ➔ ***zoom***: users are able to zoom in on elements of interest
- ➔ ***filter:*** users are able to filter out items of their interest, such as countries and years of their concern
- ➔ ***details-on-demand:*** users are able to select a year they are interested in for a specific country, and get details about its crime and immigration rates
- ➔ ***relate:*** users are able to view relationships between elements, such as map representation of EU country, that includes statistics for both crime and immigration rates and a correlation matrix
- ➔ ***history:*** our application does not support a history of operations to allow undo, redo, and apply progressive optimization
- ➔ ***extract:*** our application does not support the extraction of data in response to queries

Visual coding and interaction design

- Visual mapping. Spatial substrate

We have decided to choose ***two-dimensional spatial*** substrate (x and y axes).

- **X-axis and Y-axis (geographical space):** geographical map of European Union.

Additionally, we have added a temporal ordinal dimension to integrate the vertical scrolling with which the year change is being visualized (from 2018 to 2022). We have not counted it into the spatial dimension since it is rather controlled by interaction.

Visual coding and interaction design

- Visual mapping. Graphical elements

Our visualization uses the following graphical elements:

- ***Circles:*** used on the map to represent immigration and crime data for each country when selected.
  Circle size: encodes immigration rate.
  Circle color: encodes crime rate with a gradient from white to red.

- ***Lines:*** used in the linear plots for crime and immigration rates relation.

- ***Circles in plots:*** colored dots mark individual data points on the line plot.

- ***Selector:*** a country selector allows the user to choose which country's crime-rate trend should be displayed in the line plot.

Visual coding and interaction design

- Visual mapping. Graphical properties

The graphical properties can be applied to the elements to make them more or less perceptible to the human eye and to give them relevance in the representation. We have used next properties:

- ***Size:*** used to represent an immigration rate, where larger circle implies higher immigration rate, and smaller - lower immigration rate.

- ***Orientation:*** applied in lines of the time-series graph (the slope and direction of the line encode increases or decreases in crime/immigration rate).

- ***Color:*** used to represent crime rate, where reder color implies higher crime rate, and whiter - lower crime rate.

- ***Shapes:*** circles are chosen as the consistent shape across the map and line charts to represent data points.

Prototype

Initial drawn prototype showing story we want to tell: [prototype]()

Latest prototype using framer:

Design for values

→ *Describe the ethical values that motivate the proposed prototype.*

- ***Accessibility:*** the system created gives an access for people with different kinds of disabilities through the use of assistive technologies/certain design choices.

- ***Privacy:*** the system does not collect any user related data and ensures end-to-end security.

- ***Transparency:*** the system provides transparency, so that users can make informed choices.

- ***Sustainability & Society:*** developed system operates with keeping in mind current ecological situation.

Design for values

➔ *Comply with the ART principles.*

- **Accountability:** all of the design choices can be justified. Let's break down some of them:

  - We decided to choose this kind of visualization to ease both textual and visual comprehension for users.

  - Allowing scrolling let's users build chronologically correct chain of rates by years.

  - Including map of the EU helps visually associate countries with their size and population with proposed crime/immigration rates.

  - The white to red gradient was chosen to represent crime rates to both provide an association with higher danger level and accommodate colorblind people (due to its contrast).

  - All data was taken from official, unbiased and reliable sources.

Design for values

➔ *Comply with the ART principles.*

- **Responsibility:** the system was designed to promote trust and avoid any possible harm.
  - The topic chosen remains very sensitive nowadays and can be interpreted in unethical ways, but we made sure to provide unbiased and truthful representations of both crime and immigration rates.
  - We made sure to use reliable sources that would not reinforce any stereotypes.
  - We strongly consider the developed system as beneficial for society and general public. since it gives an access to such sensitive information to everyone.

Design for values

- ➔ *Comply with the ART principles.*
  - **Transparency:** the system was designed in a way that nothing is hidden, confusing or misleading for users.

    We strongly believe that our application makes sure that user:

    - Has an access to the data sources we have used in development
    - Clearly understands the way data is visualized thanks to the literal EU map representation
    - Understands what results are being displayed at any moment
    - Understands what interaction influence different types of outputs, that in its place stay consistent

Demo

Places
Wed Dec 10 19:42
0:00
de
93%
Google
Gemini
Gmail
Outlook
WhatsApp
YouTube
TUWEL
TISS
VoWi
Aula Global
SIGM@
HN
GitHub
Plex
http://localhost:5173
Space
adAS | Servicio de
Log in to Overleaf -
New Tab
frontend
Case study - Googl
Case study - Googl
New Document - 1.
Crime & Immigration in the EU
is there a correlation?
Are crime and immigration related in the EU, or is it just a political ploy to gain voters?
SCROLL TO BEGIN

Thank you for your attention!